

Paper / Author	Methodology Used	Key Advantage	Limitation (The Gap)
Aliyu et al., 2025	Reinforcement Learning (Proximal Policy Optimization - PPO)	Continuous dynamic adaptation & real-time personalized risk assessment (96.6% accuracy).	Models operate as "black boxes," complicating clinical interpretability, and are computationally expensive for edge devices.
Hareva et al.	IoT monitoring (vital signs, sleep, air, psychology) + Normalized Score Analysis	Holistic integration of psychological, physical, and environmental factors.	Weight factors and thresholds require further extensive clinical validation; relies on a small sample size.
bLuo et al., 2020	Comorbidity Portfolios + Machine Learning (Random Forest, SVM, LR, BPNN)	Greatly improves treatment cost prediction by balancing subpopulations.	Lacks exploration of causal relationships between comorbidities; limited sample size limits knowledge discovery.
Sathyanathan et al.	Wearable IoT sensors + K-Nearest Neighbors (KNN)	Dynamic K-value adjustments for real-time localized warnings.	Needs a longer timeframe to determine durability; faces challenges in safeguarding sensitive patient data.
Rajarajan et al.	IoT-Powered Predictive Analytics + Linear Regression (LR)	Readily interpretable baseline risk calculations with 92% accuracy.	Prone to overfitting with excessive complexity; struggles with minimizing false negative and positive rates.
Wanasinghe et al., 2024	Lightweight CNN + Multi-Feature Integration (Mel, MFCC, Chroma)	High accuracy (91.04%) across 10 disease classes with very low computational cost.	Still relies on cloud/standard processors; lacks quantization required for deployment in resource-constrained embedded edge devices.

Bhat et al., 2021	Smartphone Edge Device + Convolutional Neural Network (CNN)	Cost-effective and provide real-time predictions directly on a ubiquitous mobile device.	Exclusively relies on Peak Expiratory Flow Rates (PEFR) and basic environmental data without genomic or complex clinical personalization.
Jaimini, 2017	Multimodal IoT, EMR, and social media + Bayesian/Markov Networks	Incorporates public signals (Twitter) to explain anomalous dynamics in patient-level events.	Rely heavily on correlational (not causal) observations; event detection exhibits high error rates (e.g., 11% for cooking).
Kim et al., 2020	Long Short-Term Memory (LSTM) Networks	Efficiently learns the cumulative, long-term impacts of particulate matter over time.	Limited generalizability due to an extremely small sample size (14 patients) and relies heavily on interpolating manual data.
Liang et al.	Portable IoT sensing + Random Forest Classification	Affordably fuses both real-time environmental and physiological data (92% accuracy).	Lacks deep learning architectures for dynamic adaptation and lacks edge inference (relies on laptop/hotspot setups).
Isaac et al.	Air Quality Sensors + MATLAB Classification and Regression Trees (CART)	Localized feedback algorithms that integrate the user's travel plans with environmental data.	Highly dependent on mathematical data cleaning methods; does not process data entirely in real-time.
Bae et al., 2025	Incremental SMOTE Data Augmentation + Transfer Learning (TL)	Successfully addresses class imbalances in data-starved medical applications to prevent overfitting.	Inherent trade-off causing reductions in specificity; struggles if the original dataset is too small to generate diverse synthetic data.
Tong et al., 2020	Extreme Gradient Boosting (XGBoost) + Association Rule Mining	Automatically produces 100% explainable, rule-style rationale for	Designed only for structured, tabular data; not yet capable of handling complex deep-learning or longitudinal time-series data.

		predictions without losing accuracy.	
Dong et al., 2018	Wearable and Stationary Point-of-Care Sensors	Captures localized pediatric exposure to ozone, NO2, and VOCs efficiently.	Heavily focused on hardware and epidemiology rather than integrating advanced, individualized predictive ML frameworks.
Kadariya et al., 2019	DialogFlow NLP Chatbot (kBot) + Elasticsearch Knowledge Base	Contextualizes patient inputs for personalized medication adherence and environmental trigger warnings.	Not designed for actual medical diagnosis; relies entirely on subjective self-reporting rather than objective physiological sensors.
Our System	IoT + Genomic SNP Embeddings + 3-Speed Fusion (LSTM/XGBoost) + RAG & SHAP	Replaces population averages with biological individuality via offline, privacy-preserving AI.	Fixes biological personalization, data privacy limits, unexplainable black-box IoT inputs, and synchronous data collapse.

Component 01

Paper / Author	Methodology Used	Key Advantage	Limitation (The Gap)
Moffatt et al.	Genome-Wide Association Studies (GWAS)	Identified novel 17q21 susceptibility variants for childhood asthma.	Explains only a tiny fraction of overall asthma heritability ("missing heritability") because it ignores rare variants.
Naylor et al.	Sparse Canonical Correlation Analysis (sCCA)	Effectively reveals functionally significant expression quantitative trait loci (eQTLs).	Fails when the number of variables drastically exceeds the sample size without heavy penalization (singular variance matrix).
Perez-Garcia et al.	Elastic Net Machine Learning (EWAS)	Excellent prediction of bronchodilator drug response (BDR) (AUC 0.99).	Measures DNA methylation in whole blood only, using synchronous sampling that cannot prove real-time causality.
Corral-Acero et al.	QSP + MRI-based Cardiac Geometry (Digital Twin)	Achieves 85% dosing accuracy for ADRB1/2 targeted therapies.	High computational cost (~10 ⁴ CPU hours) limits scalable, real-time edge deployment.
Laubenbacher et al.	Neuronal Network + Wearable Motor Data	Good (78%) symptom prediction using DRD2 polymorphisms.	Limited to small-scale preclinical pilot cohorts (n=50) lacking large-scale prospective clinical validation.
Niu et al.	ODE-based Glucose-Insulin + CGM (QSP)	High (92%) HbA1c concordance for GLP-1R targeted interventions.	Relies strictly on complex differential equations; lacks decentralized, offline edge-processing capabilities.

Forno et al.	Vertical Multi-Omics Integration	Successfully combines 4 omics layers to link IL5RA to asthma exacerbations.	Massive data volume makes translation to rapid, point-of-care clinical tests extremely difficult and expensive.
Spear et al.	Admixture Mapping + GWAS	High statistical power for mapping SABA drug response in admixed populations.	Certain local ancestry peaks (e.g., 8p11) fail to reach genome-wide significance after multiple testing corrections.
Namjou et al. / Khera et al.	Polygenic Risk Scores (PRS)	Effectively stratifies patients into probabilistic disease risk subgroups.	Poor generalizability to non-European ancestries; lacks the dynamic adaptability needed for acute attack prediction.
Sun et al.	Bidirectional Mendelian Randomization (MR)	Ascertains unidirectional causality (e.g., BMI effect on asthma).	Cannot dynamically model complex, real-time gene-environment interactions over a continuous time-series.
Woodruff et al.	Microarray Transcriptomics	Defined 3 distinct Th2 gene markers (CLCA1, SERPINB2, POSTN).	Uses "bulk sequencing" which averages gene expression, masking the cellular heterogeneity vital to disease progression.
Kermani et al.	Radiomultiomics & Biomarker Clustering	Identifies and distinguishes T2-high versus T2-low phenotype clusters.	Relies on classical phenotyping criteria that result in massive overlaps (e.g., allergic vs. eosinophilic classifications).
Macowan et al.	Unsupervised Multi-Omic Factor Analysis	Captures longitudinal trajectories from early	Focuses on retrospective cohort progression rather

		wheeze to school-aged asthma.	than proactive, real-time physiological tracking.
Lin et al.	Two-Stage RF-MARS (Machine Learning)	Powerful detection of complex SNP-SNP interaction patterns.	Exclusively processes static data; lacks integration with dynamic, high-frequency wearable sensor streams.
Spick et al.	SWATH-MS Proteomics + PLS-DA	Identifies complex metabolic shifts driven by drug-disease interactions.	Generates "relevance networks" that can only show similarity scores (>0.65), completely failing to prove actual causation.
Your Opportunity	Three-Speed Multimodal Fusion (LSTM + XGBoost + Genomic Lookup)	<i>(Combine continuous IoT, ML with Genomic Risk Embeddings)</i>	Fix the lack of real-time multi-modal fusion and provide per-reading, genotype-grounded SHAP explanations.

Component 02

Component 02 Three-Speed Multimodal Fusion Architecture

(Kariyawasam L.L.N.)

Paper / Study	Methodology Used	Key Advantage	Limitation (The Gap)
AMR Multimodal Time-Series Fusion (2024)	GRU-based late fusion combining static clinical variables with multivariate physiological time-series	Cleanly separates a "static" branch from a "dynamic" branch, then merges posterior probabilities directly relevant to	Only two speeds are modelled (static vs. time-series); no formal treatment of a third, intermediate-update-rate clinical branch

		fusing genomics with IoT streams	
Multimodal Gas Sensor Fusion (thermal + chemical)	Early and late fusion of LSTM (sequence) and CNN (image) branches for gas identification	Demonstrates both early- and late-fusion variants side by side, showing late fusion preserves per-modality feature semantics	Domain is industrial gas detection, not physiological/genomic; no asynchronous update-rate handling
Cardiac MACE Risk Fusion (XGBoost, 2025)	XGBoost fusing demographics, labs, EKG parameters, and imaging-derived metrics into one tabular vector	Shows real clinical benefit of unifying many heterogeneous modalities into a single boosted-tree model with interpretable features	All inputs are harmonized into one synchronous snapshot; does not distinguish real-time vs. static vs. periodic update cadences
EmotionTFN Multi-Scale Temporal IoT Fusion (2026)	Hierarchical temporal attention fusing physiological, visual, and audio IoT streams across short/medium/long time windows	Explicitly models multiple temporal scales (0.5s–60s) with sub-200ms edge latency, proving multi-rate fusion is edge-deployable	Designed for emotion recognition, not clinical risk prediction; has no static/genomic branch or per-feature SHAP attribution
IoMT Intelligent Fusion (anomaly attribution, 2026)	Hybrid fusion of a supervised XGBoost classifier with an unsupervised LSTM autoencoder for anomaly attribution in ICU vitals	Combines a fast supervised branch with a slower unsupervised branch to distinguish genuine physiological anomalies from sensor/security faults	Two-branch fusion only; no genomic/static personalization layer and no rolling multi-hour prediction window

Pediatric ED Triage Late Fusion with Modality Dropout	Late-fusion of XGBoost (tabular vitals) and a clinical-text transformer, merged via logistic regression, trained with modality dropout	Modality dropout materially improves generalization to an unseen population relevant to generalizing across patients via leave-one-out validation	Fuses only two modalities (vitals + text); no formal three-speed separation and no continuous physiological time-series branch
ECG XGBoost Fusion Model	XGBoost fusing 12-lead ECG deep features with domain-specific hand-crafted features	Strong, well-validated example of boosted-tree fusion improving on single-source models via ablation studies	Single-domain (cardiac signal) fusion; no multi-modal asynchronous architecture or genomic conditioning
DDL-CXR Addressing Asynchronicity in Clinical Fusion (2024)	Latent diffusion model that generates an up-to-date synthetic image representation to bridge gaps between asynchronously sampled EHR time-series and imaging	Directly targets the asynchronicity problem between fast (EHR) and slow (imaging) modalities, which is the core problem Component 2 must solve	Solves asynchronicity only for two modalities (EHR + imaging) via generative interpolation, not native three-speed native fusion with SHAP-attributable features
MedPatch Confidence-Guided Multi-Stage Fusion (2025)	Confidence-guided multi-stage fusion across four clinical modalities (EHR, imaging, notes, reports), combining joint and late fusion	Notes that most prior work commits to only one fusion paradigm (joint OR late) argues for combining both, a design principle relevant to hybridizing LSTM + XGBoost + static lookup	Adds architectural complexity and computational cost; not evaluated on wearable/edge-constrained hardware
NSCLC Multimodal	Late (weighted-average) fusion of PET, CT, and clinical-	Clear empirical proof that late fusion of	Fusion is retrospective and static (imaging + one-time clinical snapshot); no real-

Radiomics Late Fusion	pathological data using Elastic Net	heterogeneous modalities materially outperforms any single modality (AUC 0.811 vs. 0.679)	time rolling-window or streaming IoT component
GeneAir (Your System)	Three-Speed Late Fusion: 60-second rolling LSTM (IoT), static SNP-embedding lookup (genomic), monthly XGBoost (clinical), unified into one interpretable feature vector	First asthma-prediction framework to formally separate and benchmark three genuinely asynchronous update rates while preserving per-feature SHAP attribution across all three streams	Requires careful validation that the late-fusion combination step does not wash out the tighter genomic personalization signal from Component 1

Component 03 Genotype-Grounded Clinical RAG Assistant

(Budara V.P.R.)

Paper / Study	Methodology Used	Key Advantage	Limitation (The Gap)
RAG Variants for Clinical Decision Support: On-Premises Deployment	Comparison of dense/sparse/hybrid retrieval and self-reflective RAG variants deployed fully on-premises for hallucination mitigation	Directly validates the on-premises deployment path (relevant to your Ollama/local-LLM approach) and explicitly flags genomics as a future multimodal retrieval direction	The genomics/proteomics extension is proposed as future work only not yet implemented or evaluated in the paper itself
PGxRAG Pharmacogenomics RAG Assistant	RAG pipeline grounding LLM responses in	Closest existing analogue to a genotype-	Focused narrowly on pharmacogenomic dosing guidance; not

	pharmacogenomics guidelines (e.g., CPIC, FDA) for drug-response queries	conditioned clinical assistant; targets the same drug-response/genomic-variant use case	integrated with a live IoT/physiological prediction pipeline or patient-specific SNP embeddings
Thyro-GenAI RAG Chatbot for Personalized Disease Management	RAG chatbot built over a curated textbook/guideline corpus, benchmarked against general-purpose commercial LLMs on personalized clinical questions	Demonstrates a domain-curated RAG chatbot can be benchmarked head-to-head against frontier LLMs for personalized query answering	Single-disease scope (thyroid) with no genomic layer; personalization is based on patient history text, not a structured genetic-variant profile
RAG in Biomedicine Survey	Survey of RAG applications spanning clinical QA (e.g., Clinfo.ai), EHR-grounded diagnostic support, and genomic variant annotation	Confirms genomic-variant interpretation via RAG is an established but still-emerging biomedical application area, supporting the novelty claim	As a survey, it catalogs existing systems rather than proposing a genotype-conditioned query pipeline itself
Retrieval-Augmented Framework for LLM-Based Clinical Decision Support	RAG pipeline harmonizing unstructured clinical narratives with structured/codified EHR data to generate therapeutic suggestions	Shows a working pattern for combining structured (codified) and unstructured (narrative) data sources in one retrieval pipeline relevant to combining GINA guidelines with genomic literature	Evaluated only on de-identified/synthetic data with preliminary plausibility checks; no formal retrieval-quality metric (e.g., RAGAS) reported
ClinicBot Guideline-Grounded RAG with Prioritized	RAG chatbot that ranks retrieved evidence by clinical authority/evidence grade rather than pure	Addresses a key weakness of naive similarity-based retrieval evidence-grade prioritization is	Evidence-grading requires a manually curated evidence hierarchy; does not condition retrieval on a

Evidence and Citations	textual similarity, with verifiable citations	directly applicable to ranking GINA guidelines above lower-quality sources	patient's individual genomic profile
GeneAir (Your System)	ChromaDB vector store indexing SNP-asthma association studies, GINA guidelines, and pharmacogenomic literature; on-premises Ollama + Mistral-7B inference with a genotype-conditioned query pipeline	First asthma decision-support RAG system to condition both retrieval and generation on a patient's germline SNP profile, evaluated via RAGAS	Retrieval quality and citation grounding must be validated against a held-out clinical query set to rule out genotype-conditioning introducing retrieval bias

Component 04 Explainable Clinical Decision Support Interface (SHAP)

(Bandara A.M.H.V.R.)

Paper / Study	Methodology Used	Key Advantage	Limitation (The Gap)
XAI Digital Twin for Early Lung Disease Detection (2026)	Seven-layer framework combining CNN/LSTM/Transformer prediction with SHAP (cross-validated against LIME) and a real-time clinical dashboard	Full pipeline from sensor acquisition to SHAP explanation to clinical dashboard structurally close to Component 4's end-to-end design	Adds a blockchain integrity layer and multiple deep architectures, increasing computational cost well beyond what commodity edge hardware can support
RAPT Real-Time Respiratory Disorder Monitoring Framework	Dual-branch CNN over respiratory audio spectrograms and metadata, with color-coded probability outputs via a Gradio interface	Demonstrates a deployable, low-latency e-Health interface for non-expert users with real-time	Interpretability is limited to raw probability outputs, not feature-level SHAP attribution; explicitly notes IoT

		respiratory classification	interoperability as an open gap
Wearable Wrist-Device Asthma Attack Early-Warning Study	Longitudinal wearable capture of heart rate, blood oxygen, and activity to build an early-warning model for pediatric asthma attacks	Validates that blood-oxygen/heart-rate/activity trends carry meaningful pre-attack signal, supporting your choice of IoT features	No explainability layer at all the model outputs a warning without attributing which physiological signal drove it, and has no genomic conditioning
Wearable Respiratory Monitoring for Opioid-Induced Depression	Wireless acoustic respiratory monitoring system detecting respiratory rate, tidal volume, and apnea in real time	Shows real-time respiratory-parameter sensing can reach clinical-grade accuracy against a ventilator reference standard	Narrow physiological scope (respiratory rate/apnea only); no ML attribution layer or per-reading explanation delivered to a clinician
XAI in Healthcare Survey (SHAP for Respiratory Diagnosis)	Survey covering SHAP-based explanation of respiratory-disease diagnoses from wearables and imaging	Confirms SHAP is an established, clinically-accepted method for explaining respiratory-condition predictions, supporting your explainability choice	Survey-level synthesis; does not itself propose a per-reading, genotype-conditioned attribution pipeline
Explainable Multi-Modal Lung Disease Detection (Respiratory Audio)	CNN-BiLSTM-Attention model explained via Grad-CAM, Integrated Gradients, and SHAP jointly	Combines multiple XAI methods (not SHAP alone) to cross-validate which features drive predictions, improving	Explanations are generated per-recording, not per continuous 60-second physiological reading, and are not linked to any drug-conflict flag

		explanation trustworthiness	
T-SHAP Temporally Stable SHAP for Real- Time Clinical Monitoring	Temporally smoothed SHAP attributions for continuous joint-level activity monitoring, framed explicitly as clinical decision support	Directly tackles a core Component 4 challenge: keeping SHAP explanations <i>stable</i> across repeated readings of the same patient, not just accurate at one instant	Applied to fall- risk/skeleton tracking, not physiological/genomic asthma risk; has no drug- conflict or prescribing- loop output
GeneAir (Your System)	SHAP TreeExplainer on the fused model generating per-reading, per-patient, genotype-conditioned attributions every 60 seconds, feeding a patient mobile alert, clinician SHAP-heatmap dashboard, and an ADRB2 drug- conflict flag	First asthma XAI system to generate genotype- conditioned, temporally continuous SHAP attributions that simultaneously drive patient self- management, drug-conflict flagging, and legally defensible clinical documentation	Must validate explanation <i>stability</i> across consecutive 60- second readings (per the T-SHAP gap above) so that attributions do not flicker in ways that undermine clinician trust